

Introduction to dtametaTMB

This vignette introduces the main functionality of the **dtametaTMB** package for meta-analysis of diagnostic test accuracy (DTA) studies. We demonstrate how to fit three commonly used models:

- the Reitsma bivariate model
- the Rutter and Gatsonis (HSROC) model
- the Hoyer model for multiple thresholds

using real datasets included in the package.

How do I load the package?

```
library(dtametaTMB)
```

Which model should I use?

The Reitsma and Rutter-and-Gatsonis models are appropriate when a single threshold per study is reported.

- The Reitsma model is most appropriate when the studies report true negatives, true positives, false negatives and false positives using the same or similar thresholds,
- whereas the Rutter and Gatsonis model is most appropriate when the studies use different thresholds.

The Hoyer model allows for multiple thresholds within studies.

How do I perform a diagnostic test meta-analysis using the bivariate binomial-model of Reitsma?

Unlike the mada package, dtametaTMB uses the binomial likelihood to model the within study distribution as recommended by The Cochrane Handbook for Systematic Reviews of Diagnostic Test Accuracy.

For sensitivity in study i , we have the number of diseased individuals testing positive: $y_{Ai} \sim \mathcal{B}(n_{Ai}, \pi_{Ai})$.

Similarly for specificity, we have the number of non-diseased individuals testing negative: $y_{Bi} \sim \mathcal{B}(n_{Bi}, \pi_{Bi})$.

At the study level, we have

$$\begin{pmatrix} \text{logit}(\pi_{Ai}) \\ \text{logit}(\pi_{Bi}) \end{pmatrix} = \begin{pmatrix} \mu_{Ai} \\ \mu_{Bi} \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} \mu_A \\ \mu_B \end{pmatrix}, \Sigma \right), \quad \text{with} \quad \Sigma = \begin{pmatrix} \sigma_A^2 & \sigma_{AB} \\ \sigma_{AB} & \sigma_B^2 \end{pmatrix}.$$

```
data("anticcp")
reitsma <- fitReitsma(data=anticcp,
                      TP=TP,
                      FP=FP,
                      FN=FN,
                      TN=TN,
                      study=study)

reitsma
#>
#> Reitsma Bivariate Random-Effects Model
#> -----
```

```

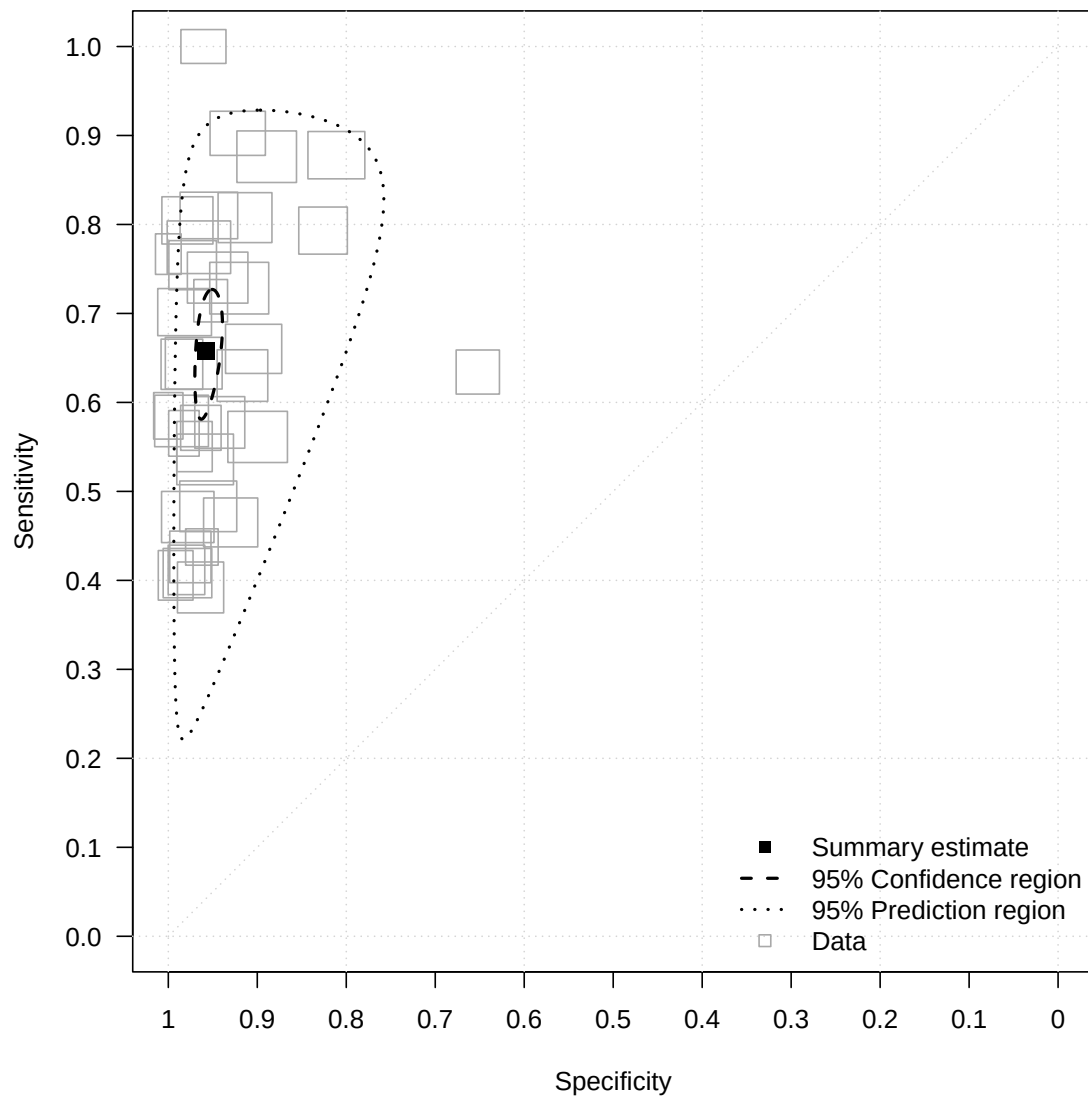
#> Number of studies: 37
#> Model fit: Converged (positive definite Hessian)
#>
#> Key parameters (logit scale):
#>   mu_A (sens): 0.653
#>   mu_B (spec): 3.109
summary(reitsma)
#> $estimates
#>           Estimate Std_Error   CI_Lower   CI_Upper
#> mu_A.sens      0.6534029 0.1274866 0.4035338 0.90327201
#> mu_B.spec      3.1089906 0.1458847 2.8230620 3.39491932
#> sigma2_A.sens  0.5426397 0.1463478 0.2558033 0.82947617
#> sigma2_B.spec  0.5717019 0.1873211 0.2045593 0.93884458
#> sigma_AB      -0.2704148 0.1198520 -0.5053204 -0.03550921
#>
#> $sensspec
#>           Estimate conflevel   CI_Lower   CI_Upper
#> sens 0.6577769          0.95 0.5995364 0.7116214
#> spec 0.9572621          0.95 0.9439094 0.9675454
#>
#> $RutterGatsonis_recovered
#>           Lambda      Theta      beta sigma2_alpha sigma2_theta
#> Estimate (recovered) 3.730684 -1.203361 0.02608605    0.573133    0.4136981

```

How do I get a summary plot of the Reitsma model?

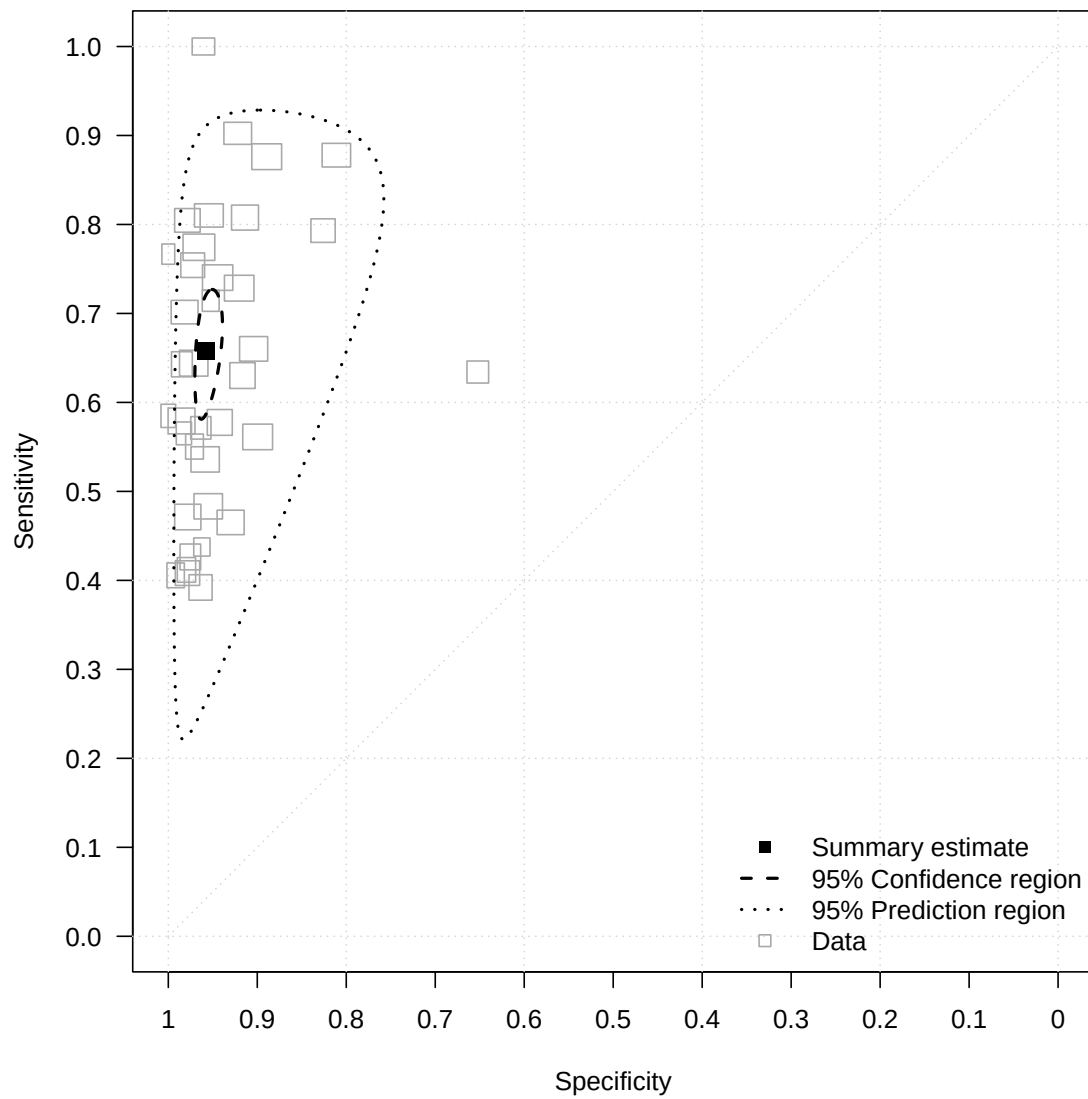
```
plot(reitsma)
```

Diagnostic Test Accuracy Meta-Analysis



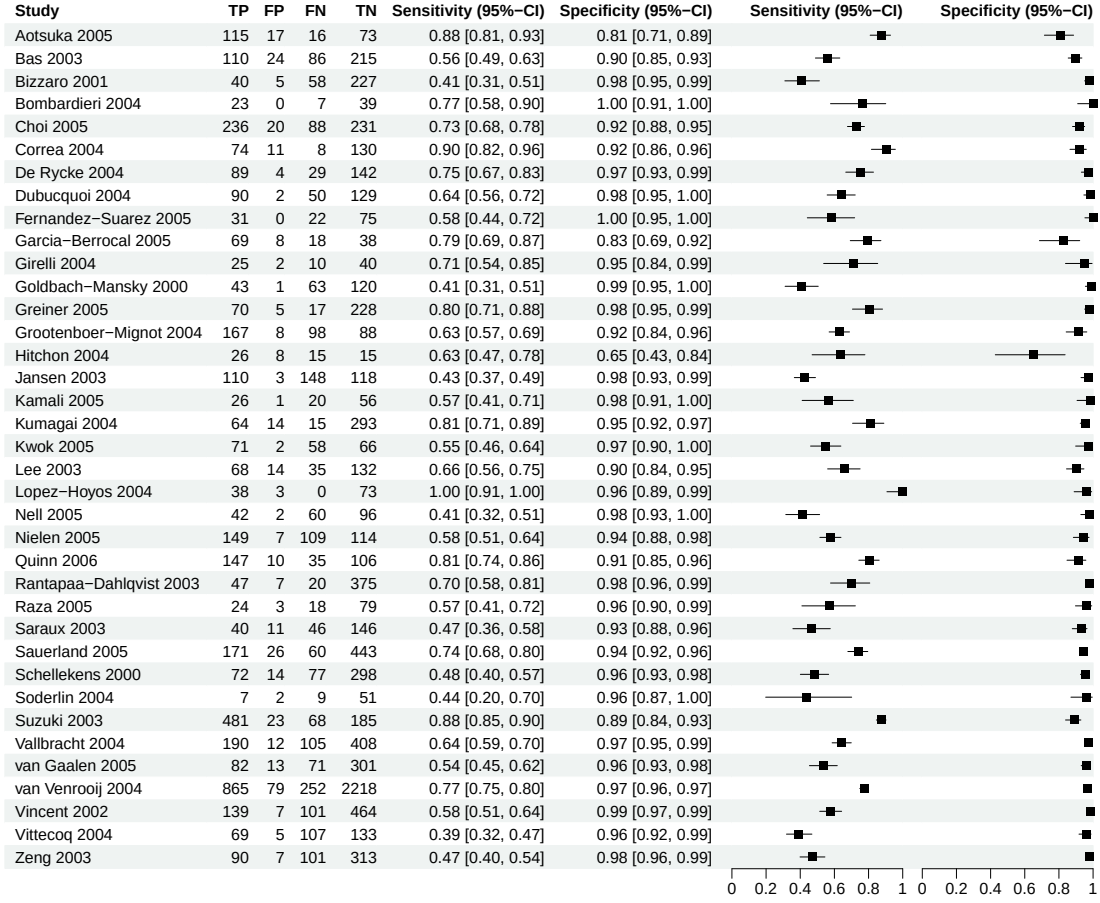
```
plot(reitsma,scale=0.01)
```

Diagnostic Test Accuracy Meta-Analysis



How do I get a coupled forest plot?

```
forest(reitsma)
```



How do I perform a diagnostic test meta-analysis using the Rutter and Gatsonis model?

The number of diseased individuals from study i who test positive is denoted by $y_{i1} \sim \mathcal{B}(n_{i1}, \pi_{i1})$.

Similarly, the number of non-diseased individuals who test positive is $y_{i2} \sim \mathcal{B}(n_{i2}, \pi_{i2})$.

At the study level, we have

$$\text{logit}(\pi_{ij}) = (\theta_i + \alpha_i x_{ij}) \exp(-\beta x_{ij}), \quad \alpha_i \sim \mathcal{N}(\Lambda, \sigma_\alpha^2), \quad \theta_i \sim \mathcal{N}(\Theta, \sigma_\theta^2),$$

where

$$x_{ij} = \begin{cases} -0.5 & \text{for non-diseased individuals,} \\ 0.5 & \text{for diseased individuals.} \end{cases}$$

```
data("RF")
ruttermatsonis <- fitRutterGatsonis(data=RF,
                                     TP=TP,
                                     FP=FP,
                                     FN=FN,
```

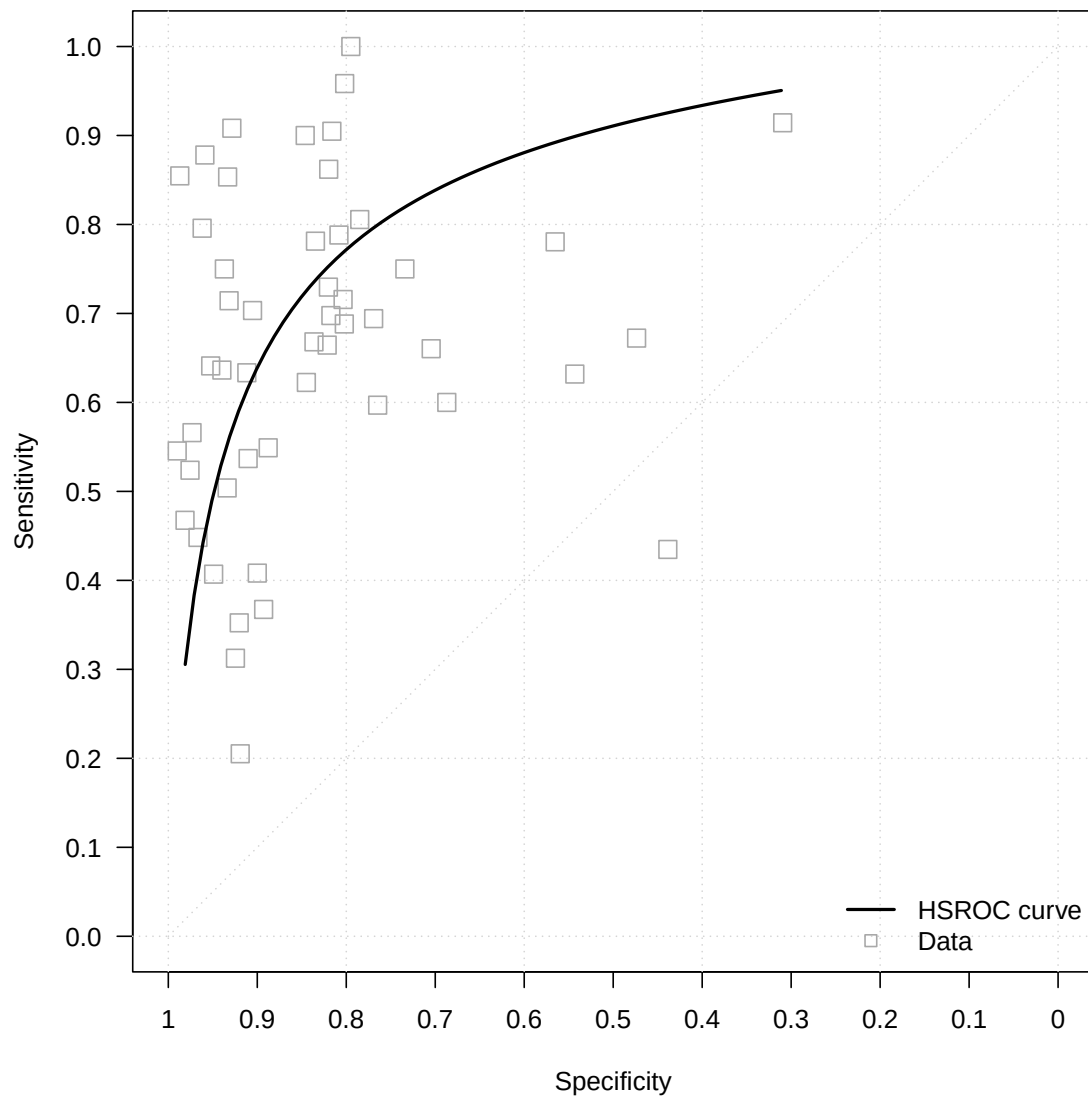
```
TN=TN,  
study=study)
```

```
ruttergatsonis  
#>  
#> Rutter & Gatsonis HSROC Model  
#> -----  
#> Number of studies: 50  
#> Model fit: Converged  
#>  
#> Parameters (HSROC scale):  
#>   Lambda: 2.602  
#>   Theta : -0.437  
#>   Beta  : 0.227  
summary(ruttergatsonis)  
#> $estimates  
#>           Estimate Std. Error  
#> Lambda      2.6015985 0.18616691  
#> Theta      -0.4370167 0.14685177  
#> beta        0.2266849 0.16236992  
#> sigma2_alpha 1.3014057 0.30455918  
#> sigma2_theta 0.5422965 0.12366256  
#> logitsens   0.8286883 0.15539440  
#> sens        0.6960775 0.03287424  
#>  
#> $sensspec  
#>      spec conflevel logitsens Std_Error CI_Lower CI_Upper      Sens  
#> 1 0.8669545      0.95 0.8286883 0.1553944 0.5241209 1.133256 0.6960775  
#>   SensCI_Lower SensCI_Upper  
#> 1    0.6281109    0.7564392  
#>  
#> $Reitsma_recovered  
#>           mu_A.sens mu_B.spec sigma2_A.sens sigma2_B.spec  sigma_AB  
#> Estimate (recovered) 0.7712238 1.946381    0.6916646    1.088408 -0.2169451
```

How do I get a summary plot of the Rutter and Gatsonis model?

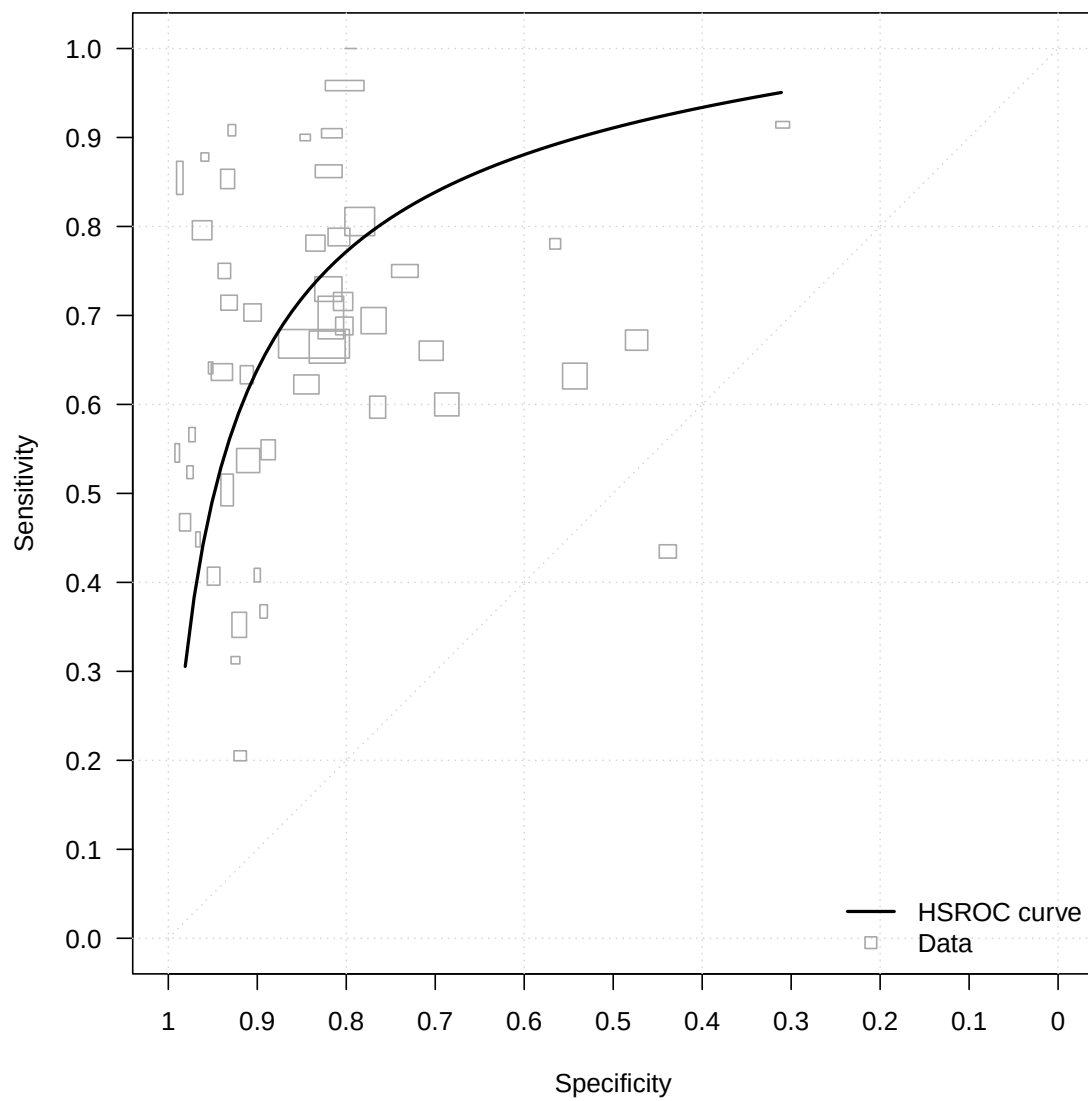
```
plot(ruttergatsonis)
```

Diagnostic Test Accuracy Meta-Analysis



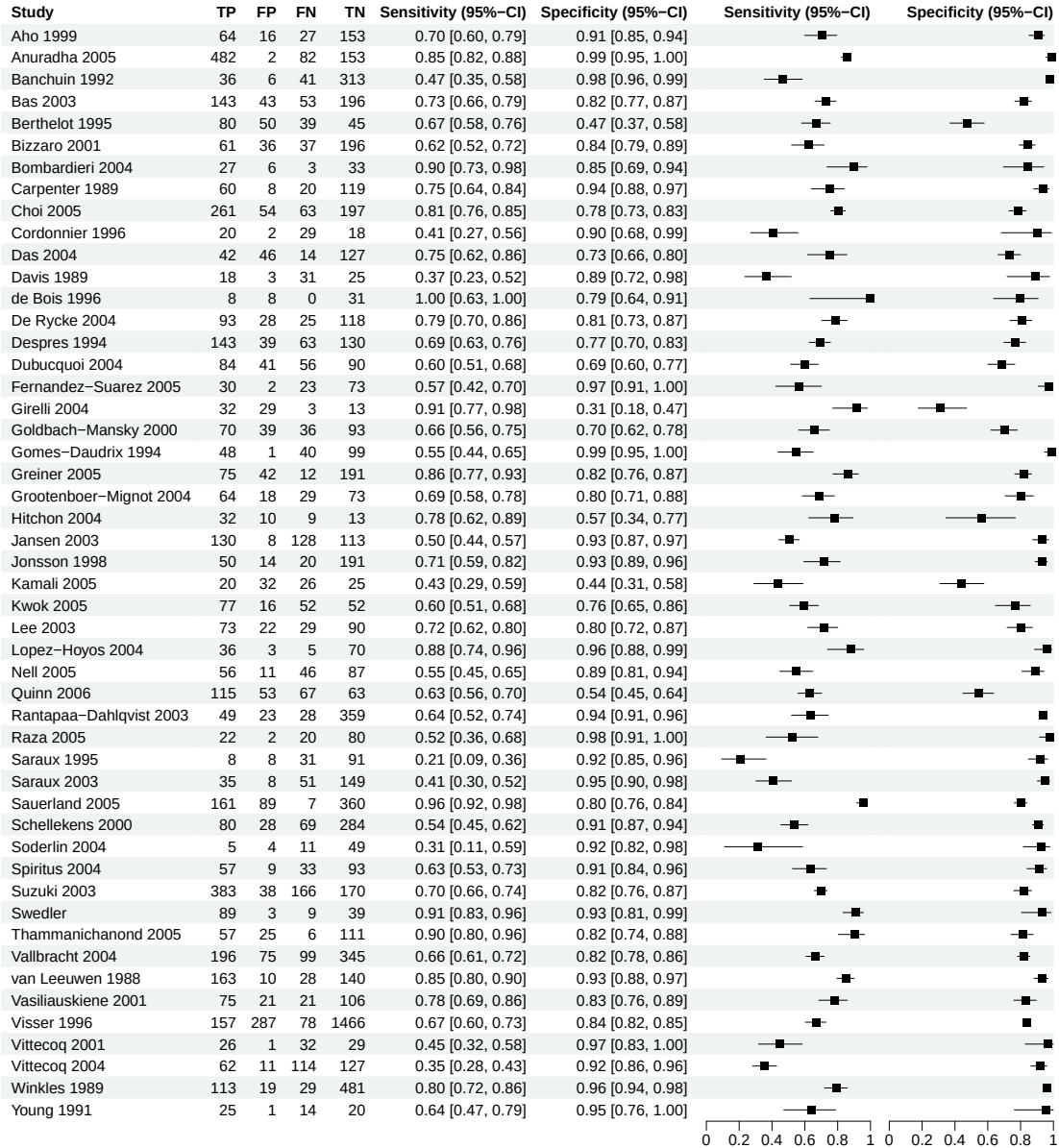
```
plot(ruttergatsonis,size="se",scale=0.01)
```

Diagnostic Test Accuracy Meta-Analysis



How do I get a coupled forest plot?

```
forest(ruttergatsonis)
```



How do I perform a diagnostic test accuracy meta-analysis with multiple thresholds per study using the Hoyer method?

The Hoyer model assumes that the diagnostic test measurement arises from a latent continuous variable Y_j , with $j = 0$ for the non-diseased individuals and $j = 1$ for the diseased, which follows a parametric survival distribution. The observed test results correspond to interval-censored observations of this latent variable due to the use of thresholds.

Which distributions does dtametaTMB support?

The latent variable Y_j is assumed to follow a location-scale family:

$$Y_j \sim F(\mu_j, \phi_j),$$

where:

- μ_j is a study-specific location parameter
- ϕ_j is a scale parameter

Common choices for F include:

Weibull distribution

$$f(y; \mu_j, \phi_j) = \mu_j \phi_j y^{\phi_j - 1} \exp(-\mu_j y^{\phi_j})$$

Log-normal distribution

$$f(y; \mu_j, \phi_j) = \frac{1}{\phi_j y} \phi \left(\frac{\log(y) - \mu_j}{\phi_j} \right)$$

Log-logistic distribution

$$f(y; \mu_j, \phi_j) = \frac{\mu_j \phi_j y^{\phi_j - 1}}{(1 + \mu_j y^{\phi_j})^2}$$

Random effects structure

To account for between-study heterogeneity and the correlation between diseased and non-diseased groups, we introduce study-specific random effects:

$$\log(y_0) = \beta_0 + \varepsilon_0 + u_{0i}, \quad \log(y_1) = \beta_1 + \varepsilon_1 + u_{1i},$$

with

$$\begin{pmatrix} u_{0i} \\ u_{1i} \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_0^2 & \rho \sigma_0 \sigma_1 \\ \rho \sigma_0 \sigma_1 & \sigma_1^2 \end{pmatrix} \right).$$

Censoring mechanism

Because only thresholded outcomes (TN, TP, FN, FP) are observed, the latent variable Y_j is subject to censoring.

- Left censoring: $(-\infty, y_k^R]$
- Interval censoring between adjacent thresholds
- Right censoring: $(y_k^L, +\infty)$

where y_k^L and y_k^R denote finite threshold values.

How do I compute sensitivities and specificities?

We can model the ROC curve by predicting sensitivities and specificities (1 - false positive rates) at several threshold values y .

Weibull distribution

$$S_1(y) = \exp(-(y \exp(-\beta_1))^{1/\lambda_1}), \quad \text{with } \beta_1 = -\log(\mu_1), \quad \lambda_1 = 1/\phi_1,$$

$$S_0(y) = \exp(-(y \exp(-\beta_0))^{1/\lambda_0}), \quad \text{with } \beta_0 = -\log(\mu_0), \quad \lambda_0 = 1/\phi_0.$$

Log-normal distribution

$$S_1(y) = 1 - \Phi\left(\frac{1}{\lambda_1}(\log(y) - \beta_1)\right), \quad \text{with } \beta_1 = \mu_1, \quad \lambda_1 = \phi_1,$$

$$S_0(y) = 1 - \Phi\left(\frac{1}{\lambda_0}(\log(y) - \beta_0)\right), \quad \text{with } \beta_0 = \mu_0, \quad \lambda_0 = \phi_0.$$

Log-logistic distribution

$$S_1(y) = 1 / \left(1 + \exp(-\beta_1/\lambda_1) y^{1/\lambda_1}\right), \quad \text{with } \beta_1 = -\phi^{-1} \log(\mu_1), \quad \lambda_1 = 1/\phi_1,$$

$$S_0(y) = 1 / \left(1 + \exp(-\beta_0/\lambda_0) y^{1/\lambda_0}\right), \quad \text{with } \beta_0 = -\phi^{-1} \log(\mu_0), \quad \lambda_0 = 1/\phi_0.$$

How do we interpret test direction? For tests where larger values indicate disease (`testdirection = "greater"`):

- $S_1(y)$ corresponds to **sensitivity** at threshold y
- $S_0(y)$ corresponds to the **false positive rate (FPR) = 1 - specificity**

For tests where smaller values indicate disease (`testdirection = "less"`):

- $S_1(y)$ corresponds to the **false negative rate (FNR) = 1 - sensitivity**
- $S_0(y)$ corresponds to **specificity**

In a nutshell

By modelling the diagnostic marker as a latent continuous outcome, the Hoyer model:

- naturally accommodates **multiple thresholds per study**
- provides a unified framework for **interval-censored diagnostic data**
- preserves the correlation between sensitivity and specificity

The following parameters have to be estimated: β_0 , β_1 , λ_0 , λ_1 , σ_0^2 , σ_1^2 , and ρ , where λ_0 and λ_1 are transformations of ϕ_0 and ϕ_1 .

To obtain initial values, it is necessary to specify a smallest (« min y) and largest (» max y) value across all studies, essentially replacing negative and positive infinity for the left and right censored intervals.

How do I fit the Hoyer model?

```
data("diabetes")
head(diabetes)
#>   study threshold TP   TN   FP   FN   D   H originally_published
#> 1     1         5.9 574 1389  682 262 836 2071                1
#> 2     2         5.0 617 1005 7735   18 635 8740                0
#> 3     2         5.1 607 1617 7123   28 635 8740                0
#> 4     2         5.2 600 2438 6302   35 635 8740                0
#> 5     2         5.3 581 3409 5331   54 635 8740                0
#> 6     2         5.4 563 4422 4318   72 635 8740                0
summary(diabetes$threshold)
#>   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
#> 3.900  5.500   5.800  5.837  6.100   7.600
```

The argument `eval_threshold` specifies the values at which sensitivities and specificities are evaluated after model fitting.

```

hoyer <- fitHoyer(data=diabetes,
                  TP=TP,
                  FP=FP,
                  FN=FN,
                  TN=TN,
                  threshold=threshold,
                  study=study,
                  dist="loglogistic",
                  testdirection="greater",
                  eval_threshold=c(5.0,5.5,6.0,6.5,7.0,7.5),
                  smallest=2,
                  largest=10)

```

```

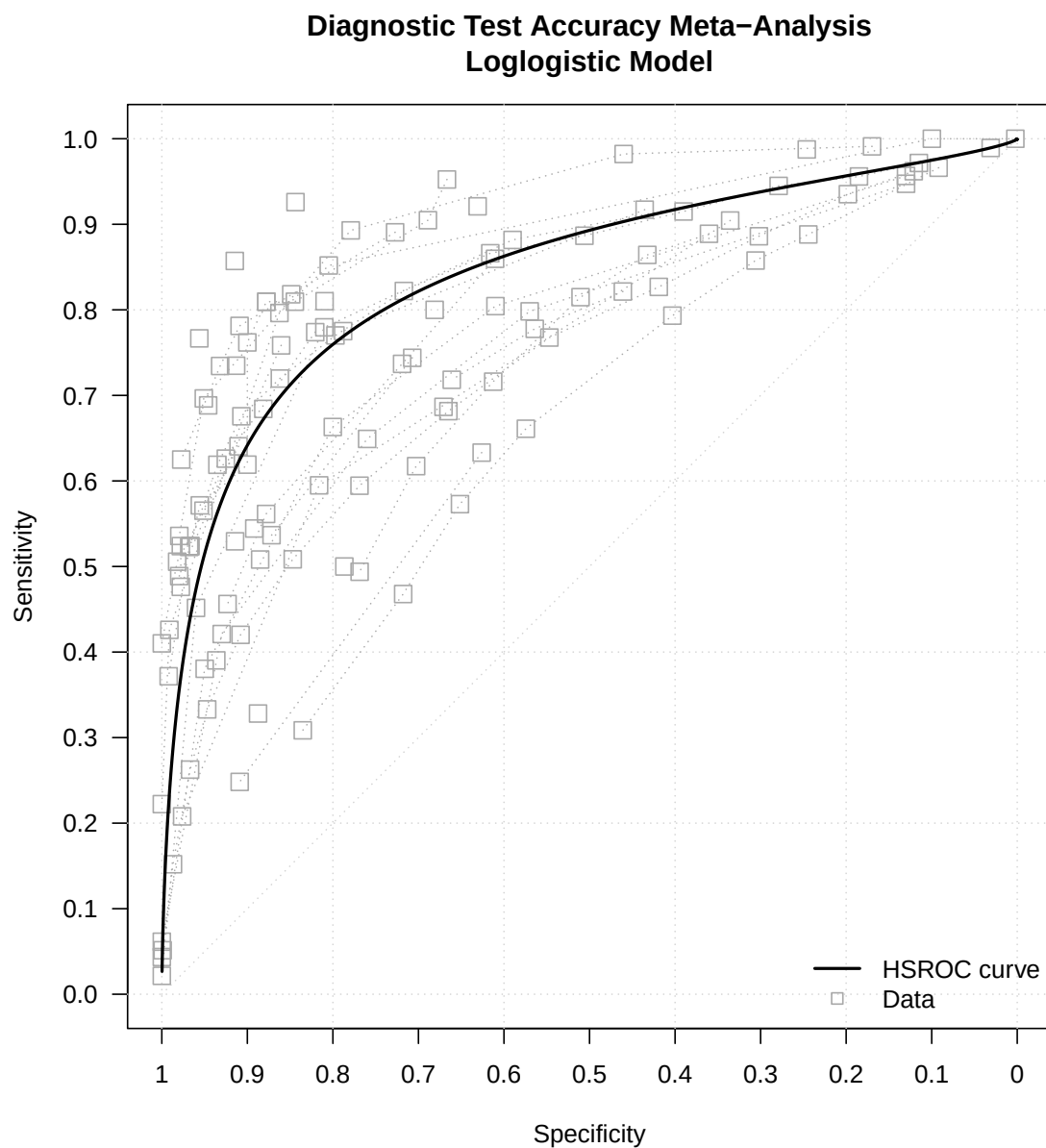
hoyer
#>
#> Hoyer AFT Model Fit
#> -----
#> Distribution : Loglogistic
#> Number of studies : 38
#> Test direction : greater
#> Optimization : converged
#> -2 negative Log Likelihood : 209322
#>
#> Use summary() for parameter estimates.
summary(hoyer)
#> $sdreport2
#>
#>           Estimate   Std. Error
#> beta0      1.705691148 0.0095630146
#> lambda0    0.039502004 0.0001857265
#> su0        0.058648679 0.0067784870
#> beta1      1.825337107 0.0145652403
#> lambda1    0.056398479 0.0009244722
#> su1        0.088320631 0.0109564611
#> rho        0.863032573 0.0454410883
#> covu0u1    0.004470412 0.0011417200
#> logitSurv1 3.828103143 0.2635035151
#> logitSurv1 2.138160760 0.2593347627
#> logitSurv1 0.595364241 0.2580790576
#> logitSurv1 -0.823870980 0.2591073990
#> logitSurv1 -2.137877556 0.2619046729
#> logitSurv1 -3.361188412 0.2660542694
#> logitSurv0 2.436667140 0.2426039991
#> logitSurv0 0.023873625 0.2420917690
#> logitSurv0 -2.178834277 0.2420878132
#> logitSurv0 -4.205129122 0.2424751080
#> logitSurv0 -6.081185122 0.2431661379
#> logitSurv0 -7.827751473 0.2440943827
#>
#> $sensspec
#>   threshold conflevel      Sens SensCI_Lower SensCI_Upper      Spec
#> 1      5.0      0.95 0.97871219  0.96482617  0.98718904 0.08041904
#> 2      5.5      0.95 0.89455725  0.83615237  0.93379170 0.49403188
#> 3      6.0      0.95 0.64459500  0.52236970  0.75048299 0.89833265
#> 4      6.5      0.95 0.30494258  0.20887630  0.42164362 0.98530044

```

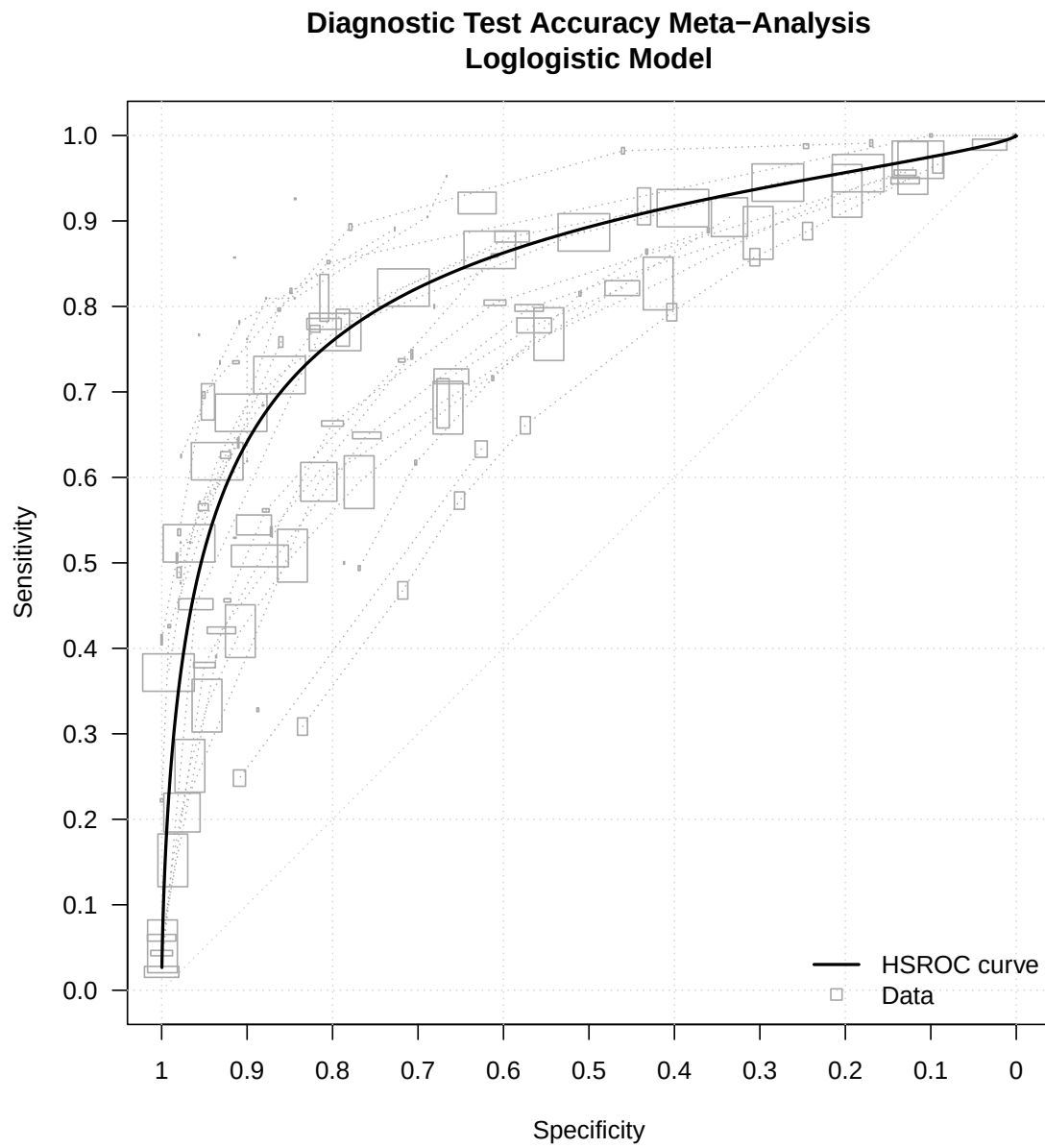
```
#> 5      7.0      0.95 0.10546947 0.06591501 0.16457780 0.99771974
#> 6      7.5      0.95 0.03353069 0.02018063 0.05521454 0.99960164
#>   SpecCI_Lower SpecCI_Upper
#> 1   0.0515556   0.1233403
#> 2   0.3779250   0.6107860
#> 3   0.8461021   0.9342152
#> 4   0.9765667   0.9908097
#> 5   0.9963326   0.9985830
#> 6   0.9993574   0.9997531
```

How do we get a summary ROC plot?

```
plot(hoyer, main="Diagnostic Test Accuracy Meta-Analysis\nLoglogistic Model")
```

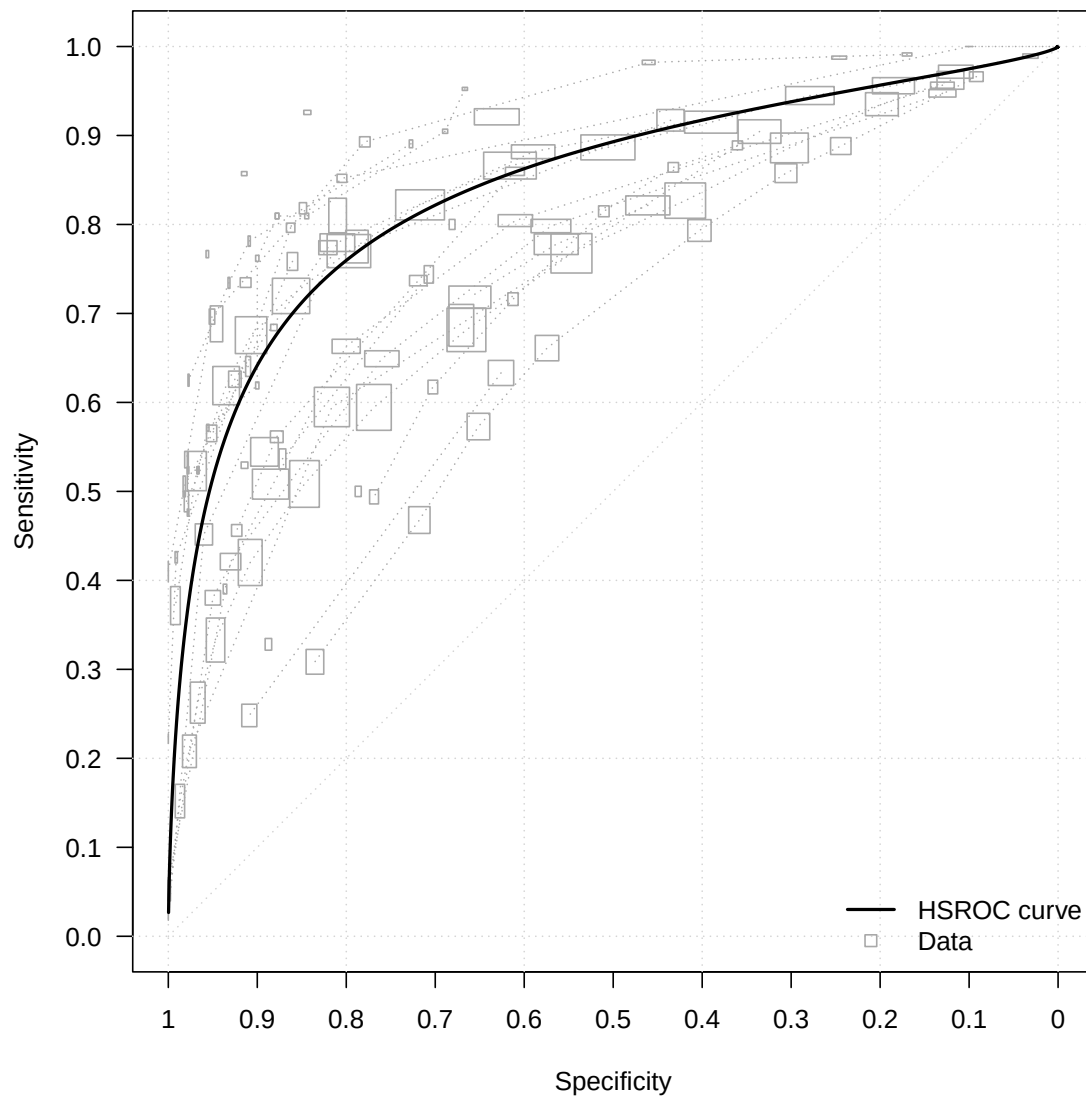


```
plot(hoyer,size="sampsize",scale=0.025,main="Diagnostic Test Accuracy Meta-Analysis\nLoglogistic Model")
```



```
plot(hoyer,size="se",scale=0.025,main="Diagnostic Test Accuracy Meta-Analysis\nLoglogistic Model")
```

Diagnostic Test Accuracy Meta-Analysis Loglogistic Model



How does `testdirection="less"` work?

We use the synthetic anaemia data set. Smaller values in haemoglobin indicate disease.

```
data("anaemia")
summary(anaemia$threshold)
#>      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
#>      7.00   8.00   10.00  10.33   13.00   14.00
```

```
less <- fitHoyer(data=anaemia,
  TP=TP,
  FP=FP,
  FN=FN,
  TN=TN,
```

```

threshold=threshold,
study=study,
dist="loglogistic",
testdirection="less",
eval_threshold=c(8,9,10,11,12,13),
smallest=5,
largest=17)

```

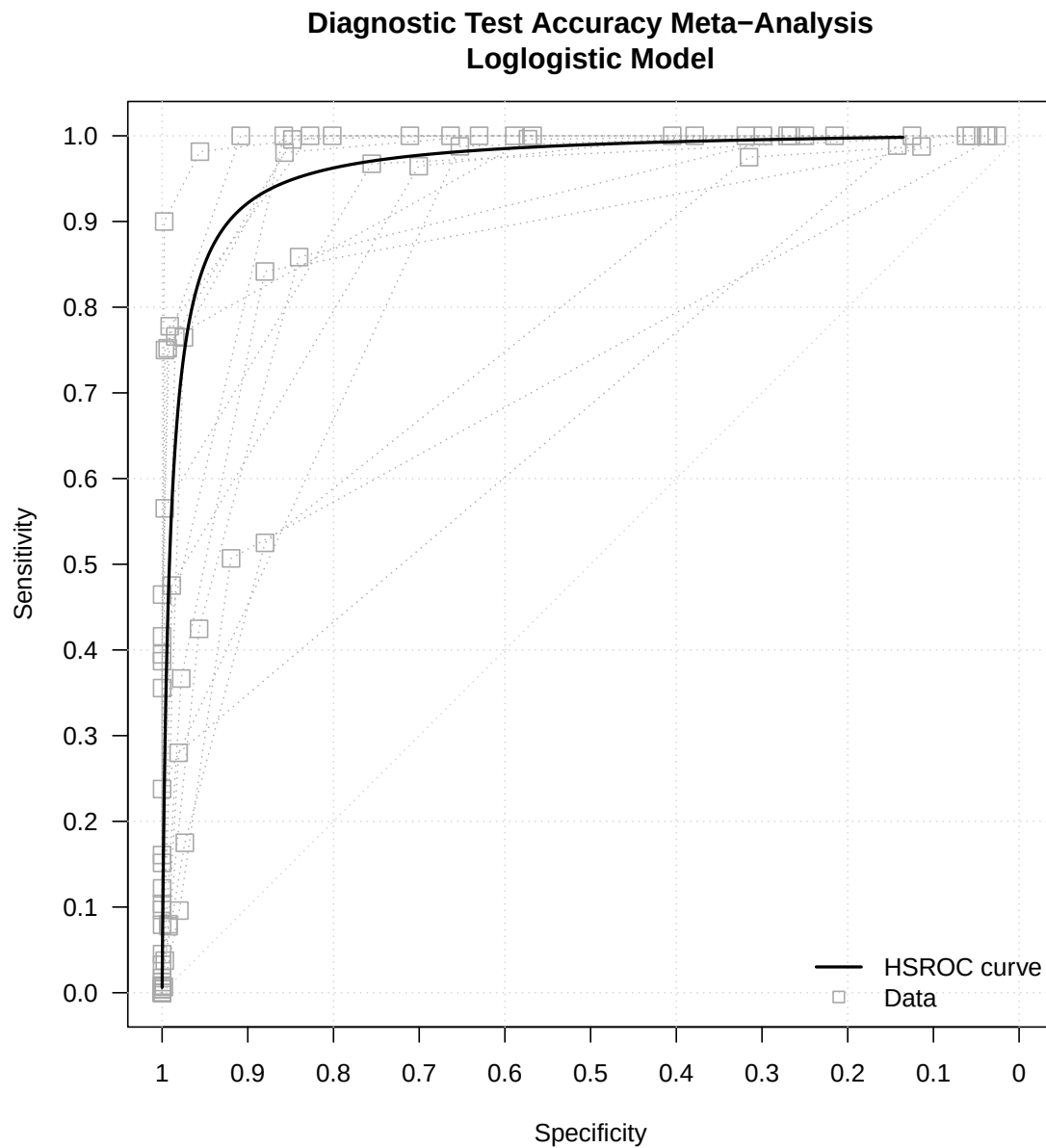
```
less
```

```

#>
#> Hoyer AFT Model Fit
#> -----
#> Distribution : Loglogistic
#> Number of studies : 15
#> Test direction : less
#> Optimization : converged
#> -2 negative Log Likelihood : 19708.9
#>
#> Use summary() for parameter estimates.
summary(less)
#> $sdreport2
#>
#>           Estimate   Std. Error
#> beta0      2.531071186 0.0169254101
#> lambda0    0.058309997 0.0007979991
#> su0        0.065266780 0.0120139561
#> beta1      2.253821209 0.0147254382
#> lambda1    0.060530034 0.0012263705
#> su1        0.056279481 0.0105351710
#> rho        -0.411758265 0.2194120812
#> covu0u1    -0.001512462 0.0010422566
#> logitSurv1 2.880878398 0.2497852789
#> logitSurv1 0.935017349 0.2438804386
#> logitSurv1 -0.805614673 0.2439346922
#> logitSurv1 -2.380207869 0.2483362272
#> logitSurv1 -3.817698831 0.2557856931
#> logitSurv1 -5.140062330 0.2652905618
#> logitSurv0 7.745320971 0.3083465399
#> logitSurv0 5.725375142 0.3001481581
#> logitSurv0 3.918472026 0.2948244780
#> logitSurv0 2.283929334 0.2917357077
#> logitSurv0 0.791708769 0.2903955841
#> logitSurv0 -0.581001083 0.2904281547
#>
#> $sensspec
#>   threshold conflevel      Sens SensCI_Lower SensCI_Upper      Spec
#> 1          8      0.95 0.05310695    0.0332319    0.08383789 0.9995674
#> 2          9      0.95 0.28190791    0.1957591    0.38769289 0.9967485
#> 3         10      0.95 0.69117423    0.5811538    0.78308198 0.9805157
#> 4         11      0.95 0.91530555    0.8691516    0.94618765 0.9075373
#> 5         12      0.95 0.97849434    0.9649861    0.98686211 0.6881981
#> 6         13      0.95 0.99417678    0.9902443    0.99652965 0.3587023
#>   SpecCI_Lower SpecCI_Upper
#> 1    0.9992086    0.9997636
#> 2    0.9941596    0.9981919

```

```
#> 3 0.9657965 0.9889730
#> 4 0.8471134 0.9456136
#> 5 0.5554075 0.7959017
#> 6 0.2404471 0.4970569
plot(less,main="Diagnostic Test Accuracy Meta-Analysis\nLoglogistic Model")
```



How does fitHoyer work internally?

Step 1: How do we restructure the data set?

```
res <- restructure(
  data = diabetes,
  TP = TP,
  FP = FP,
```

```

FN = FN,
TN = TN,
study = study,
threshold = threshold,
testdirection = "greater",
smallest = 2,
largest = 10)
res
#> $restructured
#>      study TP   TN   D   H threshold lowerB upperB events1 events0 ctype
#> 1.1      1 574 1389 836 2071      5.9      NA      5.9      262      1389      1
#> 1.2      1 574 1389 836 2071      5.9      5.9      NA      574      682      3
#> 2.1      2 617 1005 635 8740      5.0      NA      5.0       18      1005      1
#> 2.2      2 607 1617 635 8740      5.1      5.0      5.1       10      612      2
#> 2.3      2 600 2438 635 8740      5.2      5.1      5.2        7      821      2
#> 2.4      2 581 3409 635 8740      5.3      5.2      5.3       19      971      2
#> 2.5      2 563 4422 635 8740      5.4      5.3      5.4       18     1013      2
#> 2.6      2 550 5384 635 8740      5.5      5.4      5.5       13      962      2
#> 2.7      2 522 6267 635 8740      5.6      5.5      5.6       28      883      2
#> 2.8      2 489 6966 635 8740      5.7      5.6      5.7       33      699      2
#> 2.9      2 457 7534 635 8740      5.8      5.7      5.8       32      568      2
#> 2.10     2 429 7927 635 8740      5.9      5.8      5.9       28      393      2
#> 2.11     2 393 8172 635 8740      6.0      5.9      6.0       36      245      2
#> 2.12     2 332 8460 635 8740      6.2      6.0      6.2       61      288      2
#> 2.13     2 236 8670 635 8740      6.6      6.2      6.6       96      210      2
#> 2.14     2 236 8670 635 8740      6.6      6.6      NA      236       70      3
#> 3.1      3  36  998  49 1093      6.2      NA      6.2       13      998      1
#> 3.2      3  36  998  49 1093      6.2      6.2      NA       36       95      3
#> 4.1      4  16  147  41  157      6.0      NA      6.0       25      147      1
#> 4.2      4  16  147  41  157      6.0      6.0      NA       16       10      3
#> 5.1      5 644 1217 795 1503      6.1      NA      6.1      151     1217      1
#> 5.2      5 644 1217 795 1503      6.1      6.1      NA      644      286      3
#> 6.1      6 176 1286 278 2054      5.6      NA      5.6      102     1286      1
#> 6.2      6  69 1867 278 2054      6.5      5.6      6.5      107      581      2
#> 6.3      6  69 1867 278 2054      6.5      6.5      NA       69      187      3
#> 7.1      7  72  258  88  304      6.1      NA      6.1       16      258      1
#> 7.2      7  72  258  88  304      6.1      6.1      NA       72       46      3
#> 8.1      8  89 1382 115 1684      5.3      NA      5.3       26     1382      1
#> 8.2      8  72 1558 115 1684      5.5      5.3      5.5       17      176      2
#> 8.3      8  65 1602 115 1684      5.6      5.5      5.6        7       44      2
#> 8.4      8  65 1602 115 1684      5.6      5.6      NA       65       82      3
#> 9.1      9 207 2704 252 5865      5.8      NA      5.8       45     2704      1
#> 9.2      9 196 3308 252 5865      5.9      5.8      5.9       11      604      2
#> 9.3      9 181 3877 252 5865      6.0      5.9      6.0       15      569      2
#> 9.4      9 181 3877 252 5865      6.0      6.0      NA      181     1988      3
#> 10.1     10  54   1  54  451      3.9      NA      3.9        0        1      1
#> 10.2     10  54  45  54  451      4.7      3.9      4.7        0       44      2
#> 10.3     10  46 363  54  451      5.6      4.7      5.6        8      318      2
#> 10.4     10  43 389  54  451      5.7      5.6      5.7        3       26      2
#> 10.5     10  23 447  54  451      6.2      5.7      6.2       20       58      2
#> 10.6     10  12 451  54  451      6.8      6.2      6.8       11        4      2
#> 10.7     10  12 451  54  451      6.8      6.8      NA       12        0      3
#> 11.1     11 861  611 895 5050      5.0      NA      5.0       34      611      1

```

#> 11.2	11	837	1000	895	5050	5.1	5.0	5.1	24	389	2
#> 11.3	11	793	1525	895	5050	5.2	5.1	5.2	44	525	2
#> 11.4	11	740	2116	895	5050	5.3	5.2	5.3	53	591	2
#> 11.5	11	687	2762	895	5050	5.4	5.3	5.4	53	646	2
#> 11.6	11	610	3358	895	5050	5.5	5.4	5.5	77	596	2
#> 11.7	11	532	3883	895	5050	5.6	5.5	5.6	78	525	2
#> 11.8	11	455	4277	895	5050	5.7	5.6	5.7	77	394	2
#> 11.9	11	376	4585	895	5050	5.8	5.7	5.8	79	308	2
#> 11.10	11	298	4782	895	5050	5.9	5.8	5.9	78	197	2
#> 11.11	11	235	4883	895	5050	6.0	5.9	6.0	63	101	2
#> 11.12	11	136	4984	895	5050	6.2	6.0	6.2	99	101	2
#> 11.13	11	46	5045	895	5050	6.6	6.2	6.6	90	61	2
#> 11.14	11	46	5045	895	5050	6.6	6.6	NA	46	5	3
#> 12.1	12	184	5265	338	5903	5.7	NA	5.7	154	5265	1
#> 12.2	12	184	5265	338	5903	5.7	5.7	NA	184	638	3
#> 13.1	13	72	120	81	333	5.5	NA	5.5	9	120	1
#> 13.2	13	70	144	81	333	5.6	5.5	5.6	2	24	2
#> 13.3	13	66	170	81	333	5.7	5.6	5.7	4	26	2
#> 13.4	13	58	204	81	333	5.8	5.7	5.8	8	34	2
#> 13.5	13	50	234	81	333	5.9	5.8	5.9	8	30	2
#> 13.6	13	40	256	81	333	6.0	5.9	6.0	10	22	2
#> 13.7	13	40	256	81	333	6.0	6.0	NA	40	77	3
#> 14.1	14	285	155	295	1687	4.8	NA	4.8	10	155	1
#> 14.2	14	262	412	295	1687	5.1	4.8	5.1	23	257	2
#> 14.3	14	253	516	295	1687	5.2	5.1	5.2	9	104	2
#> 14.4	14	234	680	295	1687	5.3	5.2	5.3	19	164	2
#> 14.5	14	195	969	295	1687	5.5	5.3	5.5	39	289	2
#> 14.6	14	169	1099	295	1687	5.6	5.5	5.6	26	130	2
#> 14.7	14	138	1211	295	1687	5.7	5.6	5.7	31	112	2
#> 14.8	14	91	1409	295	1687	5.9	5.7	5.9	47	198	2
#> 14.9	14	91	1409	295	1687	5.9	5.9	NA	91	278	3
#> 15.1	15	108	626	114	4816	5.0	NA	5.0	6	626	1
#> 15.2	15	91	2745	114	4816	5.5	5.0	5.5	17	2119	2
#> 15.3	15	74	3660	114	4816	5.7	5.5	5.7	17	915	2
#> 15.4	15	48	4479	114	4816	6.0	5.7	6.0	26	819	2
#> 15.5	15	7	4816	114	4816	6.4	6.0	6.4	41	337	2
#> 15.6	15	7	4816	114	4816	6.4	6.4	NA	7	0	3
#> 16.1	16	184	181	186	5826	5.0	NA	5.0	2	181	1
#> 16.2	16	164	3437	186	5826	5.5	5.0	5.5	20	3256	2
#> 16.3	16	145	4719	186	5826	5.7	5.5	5.7	19	1282	2
#> 16.4	16	84	5593	186	5826	6.0	5.7	6.0	61	874	2
#> 16.5	16	4	5826	186	5826	6.4	6.0	6.4	80	233	2
#> 16.6	16	4	5826	186	5826	6.4	6.4	NA	4	0	3
#> 17.1	17	88	480	92	3692	5.0	NA	5.0	4	480	1
#> 17.2	17	74	2252	92	3692	5.5	5.0	5.5	14	1772	2
#> 17.3	17	61	2954	92	3692	5.7	5.5	5.7	13	702	2
#> 17.4	17	35	3507	92	3692	6.0	5.7	6.0	26	553	2
#> 17.5	17	4	3692	92	3692	6.4	6.0	6.4	31	185	2
#> 17.6	17	4	3692	92	3692	6.4	6.4	NA	4	0	3
#> 18.1	18	21	284	64	320	5.7	NA	5.7	43	284	1
#> 18.2	18	21	284	64	320	5.7	5.7	NA	21	36	3
#> 19.1	19	49	691	57	1132	5.1	NA	5.1	8	691	1
#> 19.2	19	42	814	57	1132	5.2	5.1	5.2	7	123	2

#> 19.3	19	32	994	57	1132	5.3	5.2	5.3	10	180	2
#> 19.4	19	26	1045	57	1132	5.4	5.3	5.4	6	51	2
#> 19.5	19	26	1045	57	1132	5.4	5.4	NA	26	87	3
#> 20.1	20	596	2062	659	6145	5.0	NA	5.0	63	2062	1
#> 20.2	20	392	5015	659	6145	5.5	5.0	5.5	204	2953	2
#> 20.3	20	137	5999	659	6145	6.0	5.5	6.0	255	984	2
#> 20.4	20	137	5999	659	6145	6.0	6.0	NA	137	146	3
#> 21.1	21	187	8562	368	9674	5.5	NA	5.5	181	8562	1
#> 21.2	21	187	8562	368	9674	5.5	5.5	NA	187	1112	3
#> 22.1	22	9	395	17	432	5.8	NA	5.8	8	395	1
#> 22.2	22	9	395	17	432	5.8	5.8	NA	9	37	3
#> 23.1	23	122	215	164	304	5.5	NA	5.5	42	215	1
#> 23.2	23	88	265	164	304	5.7	5.5	5.7	34	50	2
#> 23.3	23	88	265	164	304	5.7	5.7	NA	88	39	3
#> 24.1	24	22	129	44	164	6.1	NA	6.1	22	129	1
#> 24.2	24	22	129	44	164	6.1	6.1	NA	22	35	3
#> 25.1	25	79	217	80	882	6.6	NA	6.6	1	217	1
#> 25.2	25	79	217	80	882	6.6	6.6	NA	79	665	3
#> 26.1	26	114	203	178	223	6.9	NA	6.9	64	203	1
#> 26.2	26	90	219	178	223	7.4	6.9	7.4	24	16	2
#> 26.3	26	73	223	178	223	7.6	7.4	7.6	17	4	2
#> 26.4	26	73	223	178	223	7.6	7.6	NA	73	0	3
#> 27.1	27	135	592	178	688	5.9	NA	5.9	43	592	1
#> 27.2	27	87	674	178	688	6.5	5.9	6.5	48	82	2
#> 27.3	27	87	674	178	688	6.5	6.5	NA	87	14	3
#> 28.1	28	575	980	627	2250	5.5	NA	5.5	52	980	1
#> 28.2	28	486	1773	627	2250	6.1	5.5	6.1	89	793	2
#> 28.3	28	486	1773	627	2250	6.1	6.1	NA	486	477	3
#> 29.1	29	23	109	30	114	6.1	NA	6.1	7	109	1
#> 29.2	29	23	109	30	114	6.1	6.1	NA	23	5	3
#> 30.1	30	424	2112	616	2233	6.0	NA	6.0	192	2112	1
#> 30.2	30	424	2112	616	2233	6.0	6.0	NA	424	121	3
#> 31.1	31	338	4060	367	6436	5.7	NA	5.7	29	4060	1
#> 31.2	31	338	4060	367	6436	5.7	5.7	NA	338	2376	3
#> 32.1	32	57	32	64	44	6.0	NA	6.0	7	32	1
#> 32.2	32	50	40	64	44	6.4	6.0	6.4	7	8	2
#> 32.3	32	47	41	64	44	6.5	6.4	6.5	3	1	2
#> 32.4	32	40	43	64	44	7.0	6.5	7.0	7	2	2
#> 32.5	32	40	43	64	44	7.0	7.0	NA	40	1	3
#> 33.1	33	52	79	65	116	5.6	NA	5.6	13	79	1
#> 33.2	33	52	79	65	116	5.6	5.6	NA	52	37	3
#> 34.1	34	111	83	112	489	5.0	NA	5.0	1	83	1
#> 34.2	34	110	225	112	489	5.5	5.0	5.5	1	142	2
#> 34.3	34	100	381	112	489	6.0	5.5	6.0	10	156	2
#> 34.4	34	78	465	112	489	6.5	6.0	6.5	22	84	2
#> 34.5	34	60	479	112	489	7.0	6.5	7.0	18	14	2
#> 34.6	34	60	479	112	489	7.0	7.0	NA	60	10	3
#> 35.1	35	20	60	21	90	5.9	NA	5.9	1	60	1
#> 35.2	35	19	62	21	90	6.0	5.9	6.0	1	2	2
#> 35.3	35	17	76	21	90	6.1	6.0	6.1	2	14	2
#> 35.4	35	17	79	21	90	6.2	6.1	6.2	0	3	2
#> 35.5	35	17	79	21	90	6.3	6.2	6.3	0	0	2
#> 35.6	35	16	81	21	90	6.4	6.3	6.4	1	2	2

```

#> 35.7      35 13 81 21 90      6.5 6.4 6.5      3      0      2
#> 35.8      35 12 86 21 90      6.6 6.5 6.6      1      5      2
#> 35.9      35 11 87 21 90      6.7 6.6 6.7      1      1      2
#> 35.10     35 11 87 21 90      6.8 6.7 6.8      0      0      2
#> 35.11     35 11 88 21 90      6.9 6.8 6.9      0      1      2
#> 35.12     35 10 88 21 90      7.0 6.9 7.0      1      0      2
#> 35.13     35 10 88 21 90      7.0 7.0 NA      10      2      3
#> 36.1      36 25 243 27 288    5.9 NA 5.9      2     243      1
#> 36.2      36 25 243 27 288    5.9 5.9 NA      25      45      3
#> 37.1      37 12 268 14 293    5.9 NA 5.9      2     268      1
#> 37.2      37 12 268 14 293    5.9 5.9 NA      12      25      3
#> 38.1      38 13 260 19 295    5.9 NA 5.9      6     260      1
#> 38.2      38 13 260 19 295    5.9 5.9 NA      13      35      3
#>      lcutmean
#> 1.1      1.234050
#> 1.2      2.038769
#> 2.1      1.151293
#> 2.2      1.619339
#> 2.3      1.638950
#> 2.4      1.658183
#> 2.5      1.677053
#> 2.6      1.695574
#> 2.7      1.713757
#> 2.8      1.731616
#> 2.9      1.749162
#> 2.10     1.766405
#> 2.11     1.783356
#> 2.12     1.808154
#> 2.13     1.855809
#> 2.14     2.094827
#> 3.1      1.258848
#> 3.2      2.063567
#> 4.1      1.242453
#> 4.2      2.047172
#> 5.1      1.250718
#> 5.2      2.055437
#> 6.1      1.207957
#> 6.2      1.797284
#> 6.3      2.087194
#> 7.1      1.250718
#> 7.2      2.055437
#> 8.1      1.180427
#> 8.2      1.686227
#> 8.3      1.713757
#> 8.4      2.012676
#> 9.1      1.225503
#> 9.2      1.766405
#> 9.3      1.783356
#> 9.4      2.047172
#> 10.1     1.027062
#> 10.2     1.454270
#> 10.3     1.635165
#> 10.4     1.731616

```

```
#> 10.5 1.782508
#> 10.6 1.870736
#> 10.7 2.109754
#> 11.1 1.151293
#> 11.2 1.619339
#> 11.3 1.638950
#> 11.4 1.658183
#> 11.5 1.677053
#> 11.6 1.695574
#> 11.7 1.713757
#> 11.8 1.731616
#> 11.9 1.749162
#> 11.10 1.766405
#> 11.11 1.783356
#> 11.12 1.808154
#> 11.13 1.855809
#> 11.14 2.094827
#> 12.1 1.216807
#> 12.2 2.021526
#> 13.1 1.198948
#> 13.2 1.713757
#> 13.3 1.731616
#> 13.4 1.749162
#> 13.5 1.766405
#> 13.6 1.783356
#> 13.7 2.047172
#> 14.1 1.130882
#> 14.2 1.598928
#> 14.3 1.638950
#> 14.4 1.658183
#> 14.5 1.686227
#> 14.6 1.713757
#> 14.7 1.731616
#> 14.8 1.757709
#> 14.9 2.038769
#> 15.1 1.151293
#> 15.2 1.657093
#> 15.3 1.722607
#> 15.4 1.766113
#> 15.5 1.824029
#> 15.6 2.079442
#> 16.1 1.151293
#> 16.2 1.657093
#> 16.3 1.722607
#> 16.4 1.766113
#> 16.5 1.824029
#> 16.6 2.079442
#> 17.1 1.151293
#> 17.2 1.657093
#> 17.3 1.722607
#> 17.4 1.766113
#> 17.5 1.824029
#> 17.6 2.079442
```

```
#> 18.1 1.216807
#> 18.2 2.021526
#> 19.1 1.161194
#> 19.2 1.638950
#> 19.3 1.658183
#> 19.4 1.677053
#> 19.5 1.994492
#> 20.1 1.151293
#> 20.2 1.657093
#> 20.3 1.748254
#> 20.4 2.047172
#> 21.1 1.198948
#> 21.2 2.003667
#> 22.1 1.225503
#> 22.2 2.030222
#> 23.1 1.198948
#> 23.2 1.722607
#> 23.3 2.021526
#> 24.1 1.250718
#> 24.2 2.055437
#> 25.1 1.290108
#> 25.2 2.094827
#> 26.1 1.312334
#> 26.2 1.966501
#> 26.3 2.014814
#> 26.4 2.165367
#> 27.1 1.234050
#> 27.2 1.823377
#> 27.3 2.087194
#> 28.1 1.198948
#> 28.2 1.756518
#> 28.3 2.055437
#> 29.1 1.250718
#> 29.2 2.055437
#> 30.1 1.242453
#> 30.2 2.047172
#> 31.1 1.216807
#> 31.2 2.021526
#> 32.1 1.242453
#> 32.2 1.824029
#> 32.3 1.864050
#> 32.4 1.908856
#> 32.5 2.124248
#> 33.1 1.207957
#> 33.2 2.012676
#> 34.1 1.151293
#> 34.2 1.657093
#> 34.3 1.748254
#> 34.4 1.831781
#> 34.5 1.908856
#> 34.6 2.124248
#> 35.1 1.234050
#> 35.2 1.783356
```

```

#> 35.3 1.800024
#> 35.4 1.816419
#> 35.5 1.832549
#> 35.6 1.848424
#> 35.7 1.864050
#> 35.8 1.879436
#> 35.9 1.894589
#> 35.10 1.909515
#> 35.11 1.924222
#> 35.12 1.938716
#> 35.13 2.124248
#> 36.1 1.234050
#> 36.2 2.038769
#> 37.1 1.234050
#> 37.2 2.038769
#> 38.1 1.234050
#> 38.2 2.038769
#>
#> $original
#>      study TP   TN   FP   FN threshold   D   H testdirection   sens
#> 1      1 574 1389 682 262      5.9 836 2071      greater 0.68660287
#> 2      2 617 1005 7735 18      5.0 635 8740      greater 0.97165354
#> 3      2 607 1617 7123 28      5.1 635 8740      greater 0.95590551
#> 4      2 600 2438 6302 35      5.2 635 8740      greater 0.94488189
#> 5      2 581 3409 5331 54      5.3 635 8740      greater 0.91496063
#> 6      2 563 4422 4318 72      5.4 635 8740      greater 0.88661417
#> 7      2 550 5384 3356 85      5.5 635 8740      greater 0.86614173
#> 8      2 522 6267 2473 113     5.6 635 8740      greater 0.82204724
#> 9      2 489 6966 1774 146     5.7 635 8740      greater 0.77007874
#> 10     2 457 7534 1206 178     5.8 635 8740      greater 0.71968504
#> 11     2 429 7927 813 206      5.9 635 8740      greater 0.67559055
#> 12     2 393 8172 568 242      6.0 635 8740      greater 0.61889764
#> 13     2 332 8460 280 303      6.2 635 8740      greater 0.52283465
#> 14     2 236 8670 70 399       6.6 635 8740      greater 0.37165354
#> 15     3 36 998 95 13          6.2 49 1093      greater 0.73469388
#> 16     4 16 147 10 25          6.0 41 157       greater 0.39024390
#> 17     5 644 1217 286 151      6.1 795 1503      greater 0.81006289
#> 18     6 176 1286 768 102      5.6 278 2054      greater 0.63309353
#> 19     6 69 1867 187 209       6.5 278 2054      greater 0.24820144
#> 20     7 72 258 46 16          6.1 88 304       greater 0.81818182
#> 21     8 89 1382 302 26         5.3 115 1684      greater 0.77391304
#> 22     8 72 1558 126 43         5.5 115 1684      greater 0.62608696
#> 23     8 65 1602 82 50          5.6 115 1684      greater 0.56521739
#> 24     9 207 2704 3161 45       5.8 252 5865      greater 0.82142857
#> 25     9 196 3308 2557 56       5.9 252 5865      greater 0.77777778
#> 26     9 181 3877 1988 71       6.0 252 5865      greater 0.71825397
#> 27    10 54 1 450 0            3.9 54 451       greater 1.00000000
#> 28    10 54 45 406 0           4.7 54 451       greater 1.00000000
#> 29    10 46 363 88 8           5.6 54 451       greater 0.85185185
#> 30    10 43 389 62 11          5.7 54 451       greater 0.79629630
#> 31    10 23 447 4 31           6.2 54 451       greater 0.42592593
#> 32    10 12 451 0 42           6.8 54 451       greater 0.22222222
#> 33    11 861 611 4439 34       5.0 895 5050      greater 0.96201117

```

#> 34	11 837 1000 4050 58	5.1 895 5050	greater 0.93519553
#> 35	11 793 1525 3525 102	5.2 895 5050	greater 0.88603352
#> 36	11 740 2116 2934 155	5.3 895 5050	greater 0.82681564
#> 37	11 687 2762 2288 208	5.4 895 5050	greater 0.76759777
#> 38	11 610 3358 1692 285	5.5 895 5050	greater 0.68156425
#> 39	11 532 3883 1167 363	5.6 895 5050	greater 0.59441341
#> 40	11 455 4277 773 440	5.7 895 5050	greater 0.50837989
#> 41	11 376 4585 465 519	5.8 895 5050	greater 0.42011173
#> 42	11 298 4782 268 597	5.9 895 5050	greater 0.33296089
#> 43	11 235 4883 167 660	6.0 895 5050	greater 0.26256983
#> 44	11 136 4984 66 759	6.2 895 5050	greater 0.15195531
#> 45	11 46 5045 5 849	6.6 895 5050	greater 0.05139665
#> 46	12 184 5265 638 154	5.7 338 5903	greater 0.54437870
#> 47	13 72 120 213 9	5.5 81 333	greater 0.88888889
#> 48	13 70 144 189 11	5.6 81 333	greater 0.86419753
#> 49	13 66 170 163 15	5.7 81 333	greater 0.81481481
#> 50	13 58 204 129 23	5.8 81 333	greater 0.71604938
#> 51	13 50 234 99 31	5.9 81 333	greater 0.61728395
#> 52	13 40 256 77 41	6.0 81 333	greater 0.49382716
#> 53	14 285 155 1532 10	4.8 295 1687	greater 0.96610169
#> 54	14 262 412 1275 33	5.1 295 1687	greater 0.88813559
#> 55	14 253 516 1171 42	5.2 295 1687	greater 0.85762712
#> 56	14 234 680 1007 61	5.3 295 1687	greater 0.79322034
#> 57	14 195 969 718 100	5.5 295 1687	greater 0.66101695
#> 58	14 169 1099 588 126	5.6 295 1687	greater 0.57288136
#> 59	14 138 1211 476 157	5.7 295 1687	greater 0.46779661
#> 60	14 91 1409 278 204	5.9 295 1687	greater 0.30847458
#> 61	15 108 626 4190 6	5.0 114 4816	greater 0.94736842
#> 62	15 91 2745 2071 23	5.5 114 4816	greater 0.79824561
#> 63	15 74 3660 1156 40	5.7 114 4816	greater 0.64912281
#> 64	15 48 4479 337 66	6.0 114 4816	greater 0.42105263
#> 65	15 7 4816 0 107	6.4 114 4816	greater 0.06140351
#> 66	16 184 181 5645 2	5.0 186 5826	greater 0.98924731
#> 67	16 164 3437 2389 22	5.5 186 5826	greater 0.88172043
#> 68	16 145 4719 1107 41	5.7 186 5826	greater 0.77956989
#> 69	16 84 5593 233 102	6.0 186 5826	greater 0.45161290
#> 70	16 4 5826 0 182	6.4 186 5826	greater 0.02150538
#> 71	17 88 480 3212 4	5.0 92 3692	greater 0.95652174
#> 72	17 74 2252 1440 18	5.5 92 3692	greater 0.80434783
#> 73	17 61 2954 738 31	5.7 92 3692	greater 0.66304348
#> 74	17 35 3507 185 57	6.0 92 3692	greater 0.38043478
#> 75	17 4 3692 0 88	6.4 92 3692	greater 0.04347826
#> 76	18 21 284 36 43	5.7 64 320	greater 0.32812500
#> 77	19 49 691 441 8	5.1 57 1132	greater 0.85964912
#> 78	19 42 814 318 15	5.2 57 1132	greater 0.73684211
#> 79	19 32 994 138 25	5.3 57 1132	greater 0.56140351
#> 80	19 26 1045 87 31	5.4 57 1132	greater 0.45614035
#> 82	20 596 2062 4083 63	5.0 659 6145	greater 0.90440061
#> 81	20 392 5015 1130 267	5.5 659 6145	greater 0.59484067
#> 83	20 137 5999 146 522	6.0 659 6145	greater 0.20789074
#> 84	21 187 8562 1112 181	5.5 368 9674	greater 0.50815217
#> 85	22 9 395 37 8	5.8 17 432	greater 0.52941176
#> 86	23 122 215 89 42	5.5 164 304	greater 0.74390244

```

#> 87      23  88 265   39  76      5.7 164  304      greater 0.53658537
#> 88      24  22 129   35  22      6.1  44  164      greater 0.50000000
#> 89      25  79 217  665   1      6.6  80  882      greater 0.98750000
#> 90      26 114 203   20  64      6.9 178  223      greater 0.64044944
#> 91      26  90 219    4  88      7.4 178  223      greater 0.50561798
#> 92      26  73 223    0 105      7.6 178  223      greater 0.41011236
#> 93      27 135 592   96  43      5.9 178  688      greater 0.75842697
#> 94      27  87 674   14  91      6.5 178  688      greater 0.48876404
#> 95      28 575 980 1270  52      5.5 627 2250      greater 0.91706539
#> 96      28 486 1773  477 141      6.1 627 2250      greater 0.77511962
#> 97      29  23 109    5   7      6.1  30  114      greater 0.76666667
#> 98      30 424 2112  121 192      6.0 616 2233      greater 0.68831169
#> 99      31 338 4060 2376  29      5.7 367 6436      greater 0.92098093
#> 100     32  57  32   12   7      6.0  64  44      greater 0.89062500
#> 101     32  50  40    4  14      6.4  64  44      greater 0.78125000
#> 102     32  47  41    3  17      6.5  64  44      greater 0.73437500
#> 103     32  40  43    1  24      7.0  64  44      greater 0.62500000
#> 104     33  52  79   37  13      5.6  65  116      greater 0.80000000
#> 105     34 111  83  406   1      5.0 112  489      greater 0.99107143
#> 106     34 110 225 264   2      5.5 112  489      greater 0.98214286
#> 107     34 100 381 108  12      6.0 112  489      greater 0.89285714
#> 108     34  78 465   24  34      6.5 112  489      greater 0.69642857
#> 109     34  60 479   10  52      7.0 112  489      greater 0.53571429
#> 110     35  20  60   30   1      5.9  21  90      greater 0.95238095
#> 111     35  19  62   28   2      6.0  21  90      greater 0.90476190
#> 112     35  17  76   14   4      6.1  21  90      greater 0.80952381
#> 113     35  17  79   11   4      6.2  21  90      greater 0.80952381
#> 114     35  17  79   11   4      6.3  21  90      greater 0.80952381
#> 115     35  16  81    9   5      6.4  21  90      greater 0.76190476
#> 116     35  13  81    9   8      6.5  21  90      greater 0.61904762
#> 117     35  12  86    4   9      6.6  21  90      greater 0.57142857
#> 118     35  11  87    3  10      6.7  21  90      greater 0.52380952
#> 119     35  11  87    3  10      6.8  21  90      greater 0.52380952
#> 120     35  11  88    2  10      6.9  21  90      greater 0.52380952
#> 121     35  10  88    2  11      7.0  21  90      greater 0.47619048
#> 122     36  25 243   45   2      5.9  27  288      greater 0.92592593
#> 123     37  12 268   25   2      5.9  14  293      greater 0.85714286
#> 124     38  13 260   35   6      5.9  19  295      greater 0.68421053
#>
#>          spec          fpr
#> 1  0.670690488 0.329309512
#> 2  0.114988558 0.885011442
#> 3  0.185011442 0.814988558
#> 4  0.278947368 0.721052632
#> 5  0.390045767 0.609954233
#> 6  0.505949657 0.494050343
#> 7  0.616018307 0.383981693
#> 8  0.717048055 0.282951945
#> 9  0.797025172 0.202974828
#> 10 0.862013730 0.137986270
#> 11 0.906979405 0.093020595
#> 12 0.935011442 0.064988558
#> 13 0.967963387 0.032036613
#> 14 0.991990847 0.008009153

```

```

#> 15 0.913083257 0.086916743
#> 16 0.936305732 0.063694268
#> 17 0.809713906 0.190286094
#> 18 0.626095424 0.373904576
#> 19 0.908958130 0.091041870
#> 20 0.848684211 0.151315789
#> 21 0.820665083 0.179334917
#> 22 0.925178147 0.074821853
#> 23 0.951306413 0.048693587
#> 24 0.461040068 0.538959932
#> 25 0.564023870 0.435976130
#> 26 0.661040068 0.338959932
#> 27 0.002217295 0.997782705
#> 28 0.099778271 0.900221729
#> 29 0.804878049 0.195121951
#> 30 0.862527716 0.137472284
#> 31 0.991130820 0.008869180
#> 32 1.000000000 0.000000000
#> 33 0.120990099 0.879009901
#> 34 0.198019802 0.801980198
#> 35 0.301980198 0.698019802
#> 36 0.419009901 0.580990099
#> 37 0.546930693 0.453069307
#> 38 0.664950495 0.335049505
#> 39 0.768910891 0.231089109
#> 40 0.846930693 0.153069307
#> 41 0.907920792 0.092079208
#> 42 0.946930693 0.053069307
#> 43 0.966930693 0.033069307
#> 44 0.986930693 0.013069307
#> 45 0.999009901 0.000990099
#> 46 0.891919363 0.108080637
#> 47 0.360360360 0.639639640
#> 48 0.432432432 0.567567568
#> 49 0.510510511 0.489489489
#> 50 0.612612613 0.387387387
#> 51 0.702702703 0.297297297
#> 52 0.768768769 0.231231231
#> 53 0.091879075 0.908120925
#> 54 0.244220510 0.755779490
#> 55 0.305868405 0.694131595
#> 56 0.403082395 0.596917605
#> 57 0.574392413 0.425607587
#> 58 0.651452282 0.348547718
#> 59 0.717842324 0.282157676
#> 60 0.835210433 0.164789567
#> 61 0.129983389 0.870016611
#> 62 0.569975083 0.430024917
#> 63 0.759966777 0.240033223
#> 64 0.930024917 0.069975083
#> 65 1.000000000 0.000000000
#> 66 0.031067628 0.968932372
#> 67 0.589941641 0.410058359

```

```

#> 68 0.809989701 0.190010299
#> 69 0.960006866 0.039993134
#> 70 1.000000000 0.000000000
#> 71 0.130010834 0.869989166
#> 72 0.609967497 0.390032503
#> 73 0.800108342 0.199891658
#> 74 0.949891658 0.050108342
#> 75 1.000000000 0.000000000
#> 76 0.887500000 0.112500000
#> 77 0.610424028 0.389575972
#> 78 0.719081272 0.280918728
#> 79 0.878091873 0.121908127
#> 80 0.923144876 0.076855124
#> 82 0.335557364 0.664442636
#> 81 0.816110659 0.183889341
#> 83 0.976240846 0.023759154
#> 84 0.885052719 0.114947281
#> 85 0.914351852 0.085648148
#> 86 0.707236842 0.292763158
#> 87 0.871710526 0.128289474
#> 88 0.786585366 0.213414634
#> 89 0.246031746 0.753968254
#> 90 0.910313901 0.089686099
#> 91 0.982062780 0.017937220
#> 92 1.000000000 0.000000000
#> 93 0.860465116 0.139534884
#> 94 0.979651163 0.020348837
#> 95 0.435555556 0.564444444
#> 96 0.788000000 0.212000000
#> 97 0.956140351 0.043859649
#> 98 0.945812808 0.054187192
#> 99 0.630826600 0.369173400
#> 100 0.727272727 0.272727273
#> 101 0.909090909 0.090909091
#> 102 0.931818182 0.068181818
#> 103 0.977272727 0.022727273
#> 104 0.681034483 0.318965517
#> 105 0.169734151 0.830265849
#> 106 0.460122699 0.539877301
#> 107 0.779141104 0.220858896
#> 108 0.950920245 0.049079755
#> 109 0.979550102 0.020449898
#> 110 0.666666667 0.333333333
#> 111 0.688888889 0.311111111
#> 112 0.844444444 0.155555556
#> 113 0.877777778 0.122222222
#> 114 0.877777778 0.122222222
#> 115 0.900000000 0.100000000
#> 116 0.900000000 0.100000000
#> 117 0.955555556 0.044444444
#> 118 0.966666667 0.033333333
#> 119 0.966666667 0.033333333
#> 120 0.977777778 0.022222222

```

```
#> 121 0.9777777778 0.022222222
#> 122 0.843750000 0.156250000
#> 123 0.914675768 0.085324232
#> 124 0.881355932 0.118644068
```

Step 2: How do we get initial values?

```
init <- getInitParms(res$restructured,
                     dist="loglogistic")

init
#>
#>          beta0_init lambda0_init beta1_init lambda1_init su0_init
#> Initial value  1.681153   0.04964348   1.815414   0.08091965 0.1552818
#>
#>          su1_init covu0u1_init distcode
#> Initial value 0.1284618   0.3948626      3
```

Step 3: How do we fit the model?

```
hoyerAFT <- fitHoyerAFT(data=res,
                        init=init,
                        threshold=c(5.0,5.5,6.0,6.5,7.0,7.5))
```

```
hoyerAFT
#>
#> Hoyer AFT Model Fit
#> -----
#> Distribution : Loglogistic
#> Number of studies : 38
#> Test direction : greater
#> Optimization : converged
#> -2 negative Log Likelihood : 209322
#>
#> Use summary() for parameter estimates.
summary(hoyerAFT)
#> $sdreport2
#>
#>          Estimate Std. Error
#> beta0      1.705691148 0.0095630146
#> lambda0     0.039502004 0.0001857265
#> su0         0.058648679 0.0067784870
#> beta1      1.825337107 0.0145652403
#> lambda1     0.056398479 0.0009244722
#> su1         0.088320631 0.0109564611
#> rho         0.863032573 0.0454410883
#> covu0u1     0.004470412 0.0011417200
#> logitSurv1  3.828103143 0.2635035151
#> logitSurv1  2.138160760 0.2593347627
#> logitSurv1  0.595364241 0.2580790576
#> logitSurv1 -0.823870980 0.2591073990
#> logitSurv1 -2.137877556 0.2619046729
#> logitSurv1 -3.361188412 0.2660542694
#> logitSurv0  2.436667140 0.2426039991
#> logitSurv0  0.023873625 0.2420917690
#> logitSurv0 -2.178834277 0.2420878132
#> logitSurv0 -4.205129122 0.2424751080
```

```

#> logitSurv0 -6.081185122 0.2431661379
#> logitSurv0 -7.827751473 0.2440943827
#>
#> $sensspec
#>   threshold conflevel      Sens SensCI_Lower SensCI_Upper      Spec
#> 1      5.0      0.95 0.97871219  0.96482617  0.98718904 0.08041904
#> 2      5.5      0.95 0.89455725  0.83615237  0.93379170 0.49403188
#> 3      6.0      0.95 0.64459500  0.52236970  0.75048299 0.89833265
#> 4      6.5      0.95 0.30494258  0.20887630  0.42164362 0.98530044
#> 5      7.0      0.95 0.10546947  0.06591501  0.16457780 0.99771974
#> 6      7.5      0.95 0.03353069  0.02018063  0.05521454 0.99960164
#>   SpecCI_Lower SpecCI_Upper
#> 1   0.0515556   0.1233403
#> 2   0.3779250   0.6107860
#> 3   0.8461021   0.9342152
#> 4   0.9765667   0.9908097
#> 5   0.9963326   0.9985830
#> 6   0.9993574   0.9997531

```

References

- Reitsma, J. B., et al. (2005). Bivariate analysis of sensitivity and specificity produces informative summary measures in diagnostic reviews. *Journal of Clinical Epidemiology*, 58(10), 982–990.
- Rutter, C. M., & Gatsonis, C. A. (2001). A hierarchical regression approach to meta-analysis of diagnostic test accuracy evaluations. *Statistics in Medicine*, 20(19), 2865–2884.
- Harbord, R. M., Deeks, J. J., Egger, M., Whiting, P., & Sterne, J. A. C. (2007). A unification of models for meta-analysis of diagnostic accuracy studies. *Biostatistics*, 8(2), 239–251.
- Riley, R. D., Ensor, J., Jackson, D., & Burke, D. L. (2018). Deriving percentage study weights in multi-parameter meta-analysis models. *Statistical Methods in Medical Research*, 27(10), 2885–2905.
- Hoyer, A., Hirt, S., Kuss, O. (2018). Meta-analysis of full ROC curves using bivariate time-to-event models for interval-censored data. *Research Synthesis Methods*, 9(1).
- Deeks, J. J., Bossuyt, P. M., Leeflang, M. M., & Takwoingi, Y. (editors) (2023). *Cochrane Handbook for Systematic Reviews of Diagnostic Test Accuracy*. Version 2.0 (updated July 2023). Cochrane. Available from <https://training.cochrane.org/handbook-diagnostic-test-accuracy/current>